Name:

ID:

ECO101- Introduction to Microeconomics

Section: 3 **Quiz: 1** **Marks: 20** **Time: 25 minutes**

Question 1

	Time required to produce one **T-shirt**	Time required to produce one **Trouser**
China	60/ 20 mins = 3	60/ 5 mins 12
Bangladesh	60/ 30 mins = 2	60/ 15 mins = 4

a. What is the opportunity cost of China to produce 5 t-shirts? **(4 Marks)**

b. What is the opportunity cost of Bangladesh to produce 10 trousers? **(4 Marks)**

c. Which country has an absolute advantage in producing Trousers? **(4 Marks)**

KFU - ECO101 PRACTICE QUESTION

1.a) Suppose the Demand for pizza rises ~~by~~ from 1000 to 2000 pizza a day in response to a change in the price of Burger from 250tk to 400TK. Calculate the cross price elasticity of demand for pizza with respect to the price of burger.

Here,

$Q_{Pizza_1} = 1000$

$Q_{Pizza_2} = 2000$

$P_{Burger_1} = 250$ TK

$P_{Burger_2} = 400$ TK

∴ Cross Price Elasticity

$$= \frac{\dfrac{2000-1000}{\frac{2000+1000}{2}}}{\dfrac{400-250}{\frac{250+400}{2}}}$$

$$= \frac{\dfrac{\cancel{1000}^{2}}{\cancel{1500}_{3}}}{\dfrac{\cancel{150}^{6}}{\cancel{325}_{13}}}$$

$$= \frac{2}{3} \times \frac{13}{6} = 1.44$$

∴ Demand for pizza is Elastic and Since elasticity is positive, Burgers and pizzas are substitute.

b) If your income increases from 30,000 TK to 40,000 TK a month, your demand for milk powder rises from 300 to 500 gms and demand for tea falls from 100 cups to 75 cups a month. Calculate the income elasticity of demand for milk powder and tea. Which is normal and which is inferior good?

$I_1 = 30,000$ TK

$I_2 = 40,000$ TK

$Q_{Milk_1} = 300$ gms

$Q_{milk_2} = 500$ gms

$Q_{Tea_1} = 100$ cups

$Q_{Tea_2} = 75$ cups.

Income Elasticity of Demand for milk powder

$$= \frac{\dfrac{Q_{milk_2} - Q_{milk_1}}{\frac{Q_{milk_2} + Q_{milk_1}}{2}}}{\dfrac{I_2 - I_1}{\frac{I_2 + I_1}{2}}}$$

$$= \frac{\dfrac{500-300}{\frac{500+300}{2}}}{\dfrac{40,000-30,000}{\frac{40,000+30000}{2}}}$$

$= 1.75$ (Since Elasticity is positive milk powder is a normal good)

∴ Income Elasticity of Demand for tea

$$= \frac{\dfrac{Q_{Tea_2} - Q_{Tea_1}}{\frac{Q_{Tea_2} + Q_{Tea_1}}{2}}}{\dfrac{I_2 - I_1}{I_2 + I_1}} = \frac{\dfrac{75-100}{\frac{75+100}{2}}}{\dfrac{40,000-30,000}{\frac{40,000+30,000}{2}}} = \frac{-175}{175} = -1$$

[Since the elasticity is negative tea is an inferior good]

Practice sheet 20301070

Chapter 7: Welfare and Efficiency [Practice problem]

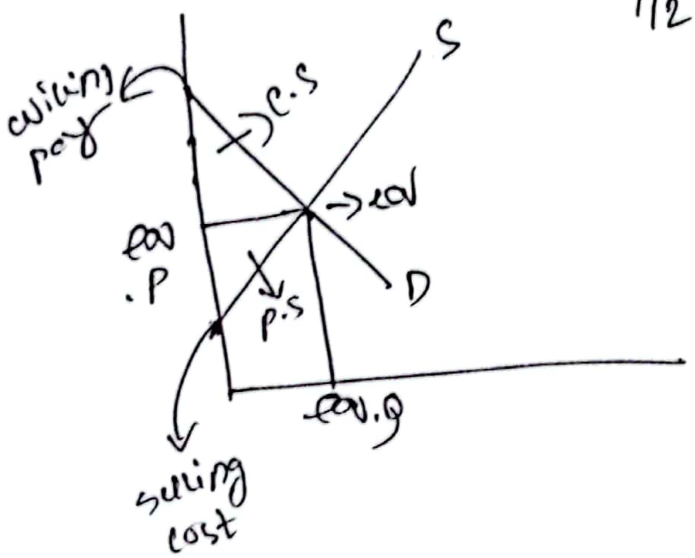
The Demand and Supply of Grameenphone SIM in Bangladesh is given by the following equations.

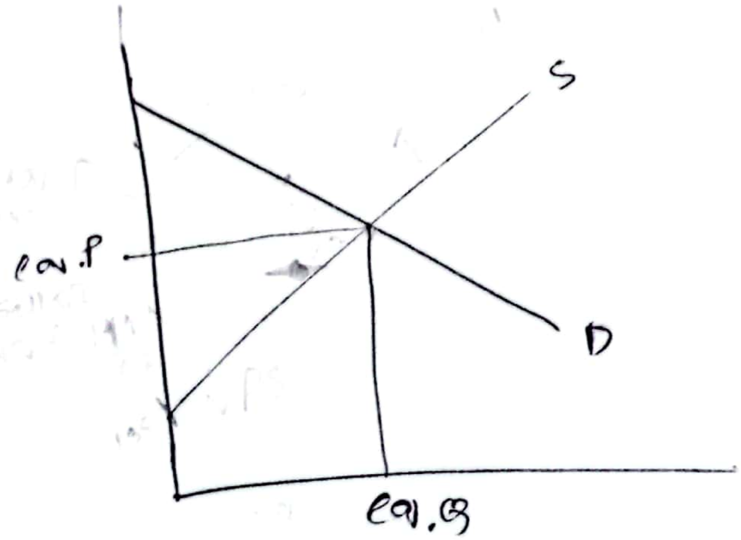
P = -16QD + 300 , Qd is quantity demanded and P is the price per SIM

P= 21Qs + 135 , Qs is quantity supplied and P is the price per SIM

a) Calculate the **Consumer Surplus** and **Producer Surplus**
b) In the recent Budget, the finance minister has proposed additional tax on the cellular services. If a 7tk tax is imposed on sellers, calculate the after tax **Consumer and Producer surplus**.
c) Compute the **Government tax revenue** 4.77 X 7
d) Calculate the **Deadweight loss** to the society after the imposition of tax. Draw an appropriate graph to show the deadweight loss area.

1/2 (w.p - eq.P) x eq.Q.





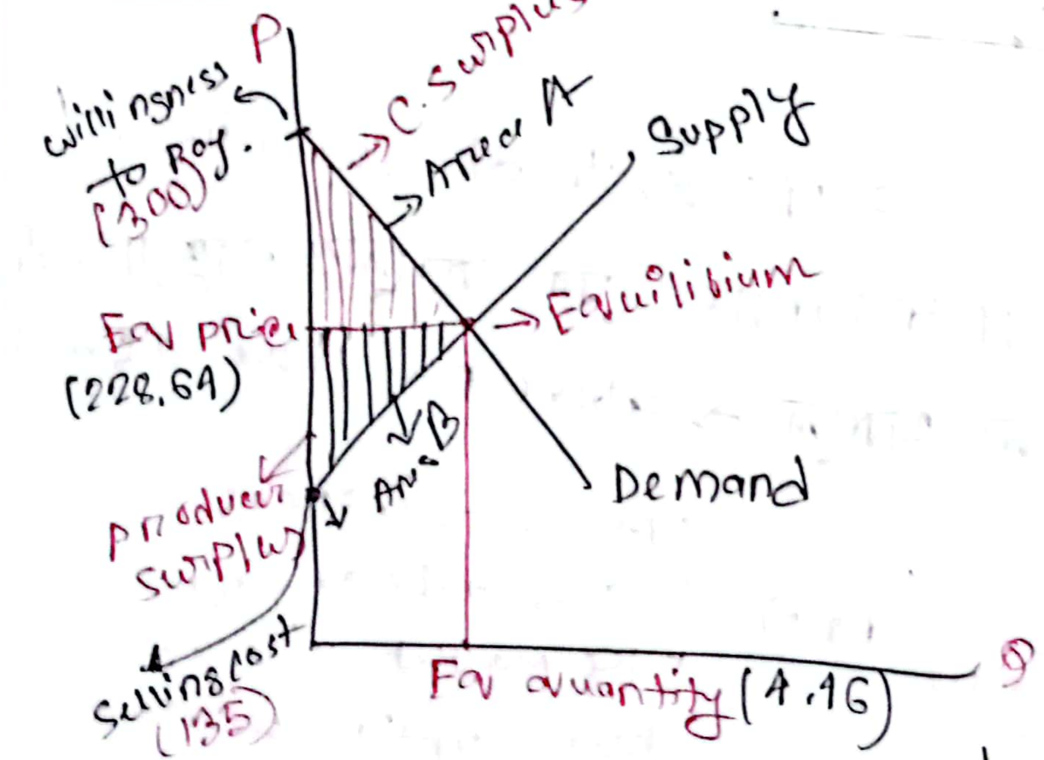
4 Question → ans any three

Ans: (a)

$P = -16Q_D + 300$ → Demand — (i)
$P = 21Q_S + 135$ → supply — (ii)

→ Demand eq for (−) slope, Relation with price (− slope with price)
→ (+ relation is supply)

Graph:



we need Actual price, willness to pay for the consumer
Eq price, quantity, selling cost of producer.

Here, $P(\text{Demand}) = -16Q_D + 300$ —(i)

$P(\text{supply}) = 21Q_S + 135$ —(ii)

at, Equilibrium, $Q_d = Q_s$ and there is equilibrium price. so, (i) = (ii).

From eq (i), (ii) ⇒

$-16Q_d + 300 = 21Q_s + 135$

$-16Q + 300 = 21Q + 135$ [∵ $Q_d = Q_s = Q$]

$\Rightarrow Q = 37Q = 165$; $\boxed{Q = 4.46}$ → Quantity can be fraction. (No need to change fraction).

putting $Q = 4.46$ in eq (ii) ⇒

$P = (21 \times 4.46 + 135) = \boxed{228.64}$

williness to pay can be found by subsititing $Q_d = 0$ in eq (i)

$\Rightarrow P = 16 \times 0 + 300 = 300$

seller's production cost can be found by substituting $Q_s = 0$ in eq (ii)

$P.P = 21 \times 0 + 135 = \boxed{135}$

$CS = \text{Area } A$

$= \frac{1}{2}(300 - 228.64) \times 4.16$

$= \boxed{159.22}$ (Ans)

$PS = \text{Area } B = \frac{1}{2}(228.64 - 135) \times 4.16$

$= \boxed{208.82}$ (Ans)

Ex: total surplus ও আসতে পারে।

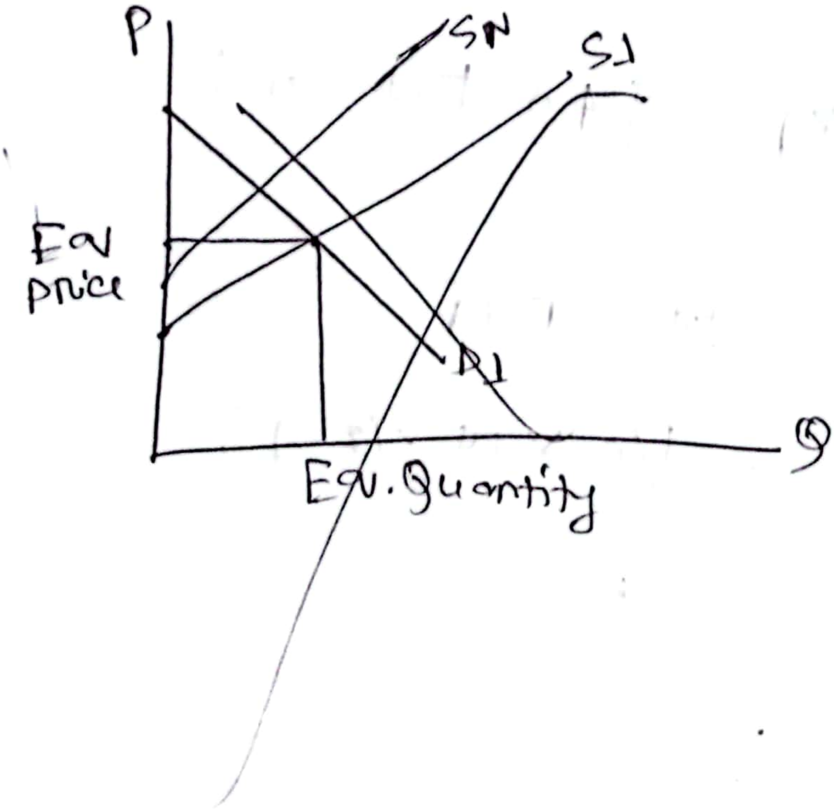
Rest of the part of practice sheet

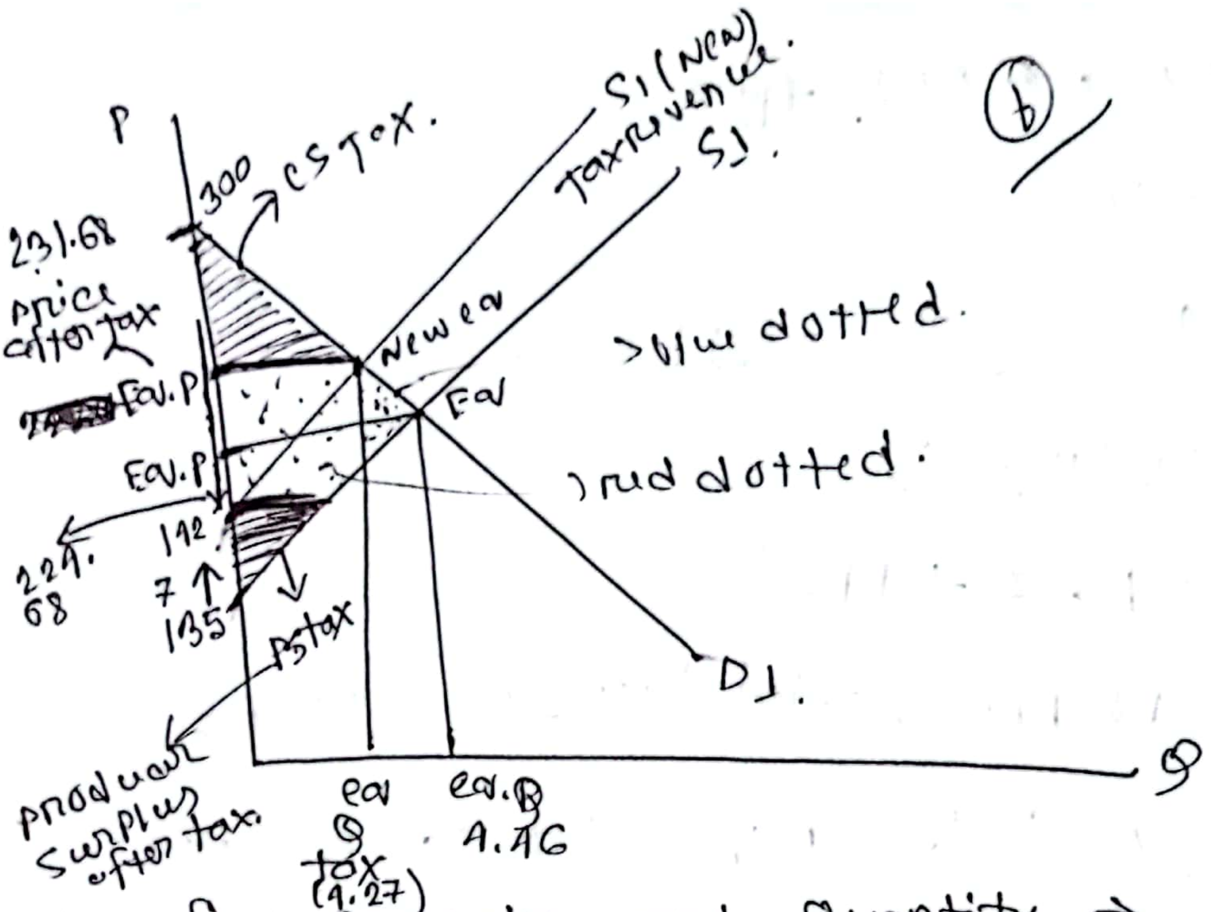
Chapter 07

(6) find out after consumer surplus

⇒

previous graph:





Now find eq. price, and quantity ⇒

135 increase by 7. So, New supply curve 142

As, the tax is imposed on sellers, the supply curve shifts leftward and the new supply equation is,.

$$P = 21Q_s + (135+7) \quad \text{——— (III)}$$

After tax eqm occurs at the intersections of New supply curve and old demand curve from (I) and (III).

$-16Q_d + 300 = 21Q_s + 142$

$\Rightarrow 37Q = 158$

$Q = 4.27 \quad [Q_s = Q_d]$

from eq (iii) $\Rightarrow$

$P = (21 \times 4.27 + 142)$

$= 231.68$ (price) after tax. [Buyers pay]

As, they need to pay 7 taka in tax $\Rightarrow$

so, $(231.68 - 7) = 224.68$

so, sellers take home income 224.68

After tax, $CS = 1/2\,(300 - 231.68) \times 4.27$

$= \boxed{}$

After tax PS $= 1/2\,(224.68 - 135) \times 4.27$

$= \boxed{}$

(c) Dotted area is tax revenue.

(d) The area that doesn't go anywhere
(Deadweight loss ⇒ blue dotted area)

(c)
Tax revenue is $=(7+k \times 9.27)$

$= \boxed{}$

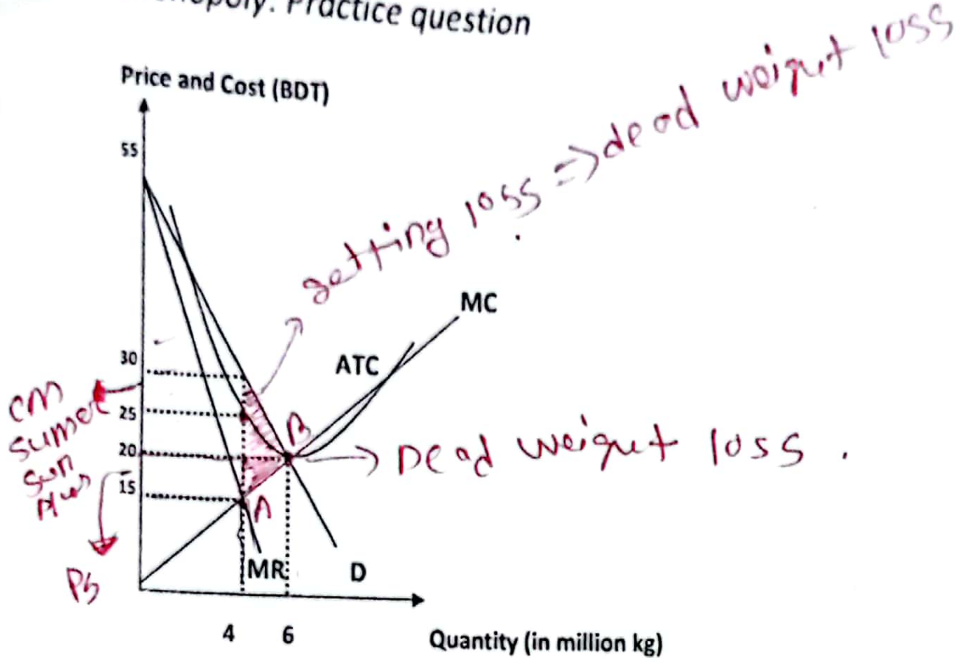
(d) Dead. w loss $= \frac{1}{2}(231.68 - 224.58) \times \cancel{4.527}$ $(1.96 - 9.27)$

$= \boxed{}$

——X——

darin

Monopoly: Practice question



a) What is the ***profit maximizing*** price and output if the firm is a ***monopoly***? *(2 marks)*

Q_M = 4 million kg P_M= BDT 30.

b) What is the ***profit maximizing*** price and output if the firm is a ***perfectly competitive*** one? *(2 marks)*

Q_{PC} = 6 million kg P_{PC}= BDT 20.

c) Under ***monopoly***, what is the economic ***profit/loss***? *(Show your calculations) (2 marks)*

Profit = Q(P-C) = 4* (30-25) = BDT 20 million*

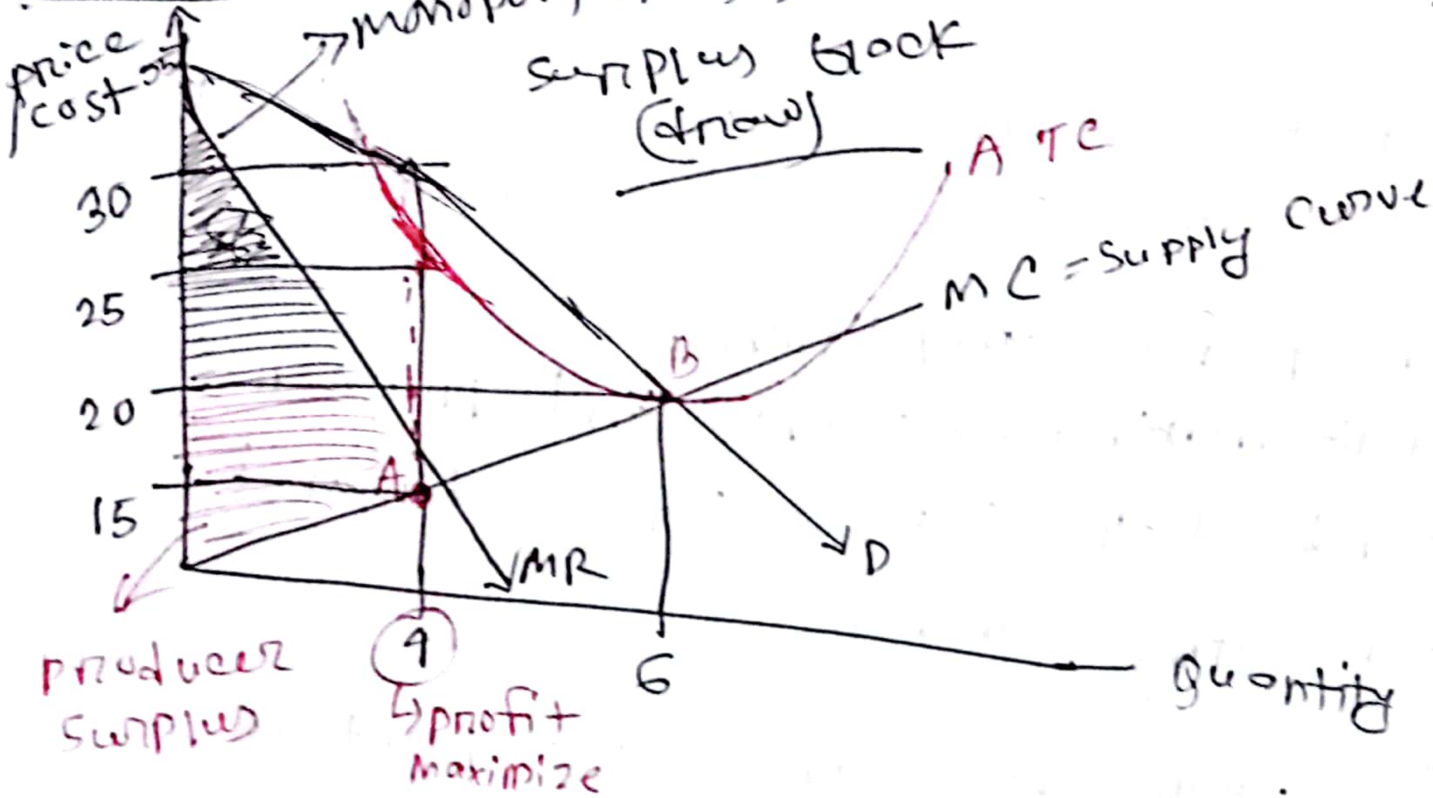
d) Under ***perfect competition***, what is the economic profit/loss? *(1 marks)*

Zero since P=ATC

e) Mark down the dead weight loss incurred between these two market structures.

Monopoly

Practice 29.09.24



(a) for a monopoly firm profit is maximum where MR = MC and the corresponding price is derived from

The demand curve since MR = MC at point A, so profit maximizing output is 4.

for output = 4, The corresponding price is 30.

Monopoly
profit Maximizing output = 4 million.

(b)

perfect comp profit = 0,
also, demand, supply → equilibrium

for perfect competition,
profit maximizing output and price is found at the intersection of demand and supply curve.

Since, demand = supply at point B
So profit maximizing output is 6 milim
and price is 20

(c) monopoly always

we know, profit (monopoly) = [to.

or, {price - Avg.

$= (30-25)$ TK

$= 5$ TK (per unit)

monopoly at point A, revenue is 30 each unit, And ATC is at 25.

Total profit $= (5 \times 4) = 20$ million TK.

(d) proofit$_{pc}$ $= (P - ATC) = 20 - 20 = 0$

So, per unit proofit $= 0$

Total proofit $= (0 \times 6)$ million

$= 0$

can be ask also,

(c) calculate the total surpluses

~~debit~~ loss = loss in dead weight

perfect monoply - total supplies বাড়লেও dead weight loss নাই।

monopoly insufficient cz dead weight loss

explain perfect competitive vs monopoly

=> P.C → more efficient than monopoly কারণ, monopoly তে dead weight loss আছে, কিন্তু PC তে নেই, কারণ surplus circle এ total surplus বেশি হয় যুক্ত।

so, dead weight loss $= \frac{1}{2} \times (30-15)(6-4)$

$= 15$

Jannatun

Perfect Competition Questions

Question 1

The market for raw beef is a perfectly competitive industry where all the firms are identical with identical cost curves. Furthermore, suppose that a representative beef supplier's total cost is given by the equation $TC = 200 + 5q^2 + 2q$ where q is the quantity of output produced by the firm. Additionally, the market demand for beef is given by the equation $P = 3000 - 6Q$ where Q is the market quantity and the market supply curve is given by the equation $P = 200 + Q$.

a. What is the equilibrium quantity and price in this market given this information?

To find the equilibrium set market demand equal to market supply: $3000 - 6Q = 200 + Q$. Solving for Q, you get $Q = 400$. Plugging 300 back into either the market demand curve or the market supply curve you get $P = 600$tk.

b. The firm's MC equation based upon its TC equation is $MC = 10q + 2$. Given this information and your answer in part (a), what is the firm's profit maximizing level of production, total revenue, total cost and profit at this market equilibrium? Is this a short-run or long-run equilibrium? Explain your answer.

From part (a) you know the equilibrium market price is 600tk. You also know that the firm profit maximizes by producing that level of output where MR = MC. Since the equilibrium market price is the firm's marginal revenue you know that MR = 600tk. Setting MR = MC gives you $600 = 10q + 2$, or $q = 59.8$. Thus, the profit maximizing level of output for the firm is 59.8 units when the price is $400 per unit.

Using this information it is easy to find total revenue as the price times the quantity: TR = (600tk per unit)(59.8 units) = 35880tk.

Total cost is found by substituting $q = 59.8$ into the TC equation: TC = 18199.8tk.

Profit is the difference between TR and TC: Profit = TR – TC = 35880 – 18199.8 = 17680.2tk

Since profit is not equal to zero this cannot be a long-run equilibrium situation: it must be a short-run equilibrium situation.

c. Given your answer in part (b), what do you anticipate will happen in this market in the long-run?

Since there is a positive economic profit in the short run, there should be entry of firms in the long-run resulting in an increase in the market quantity, a decrease in the market price, and firms in the industry earning zero economic profit.